Classification: Basic Concepts & Decision Trees

Data Mining

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Catching tax-evasion

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Tax-return data for year 2011

A new tax return for 2012
Is this a cheating tax return?

Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?

An instance of the classification problem: learn a method for discriminating between records of different classes (cheaters vs non-cheaters)

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What is classification?

- **Classification** is the task of ***learning a target function f*** that maps attribute set **x** to one of the predefined class labels **y**

categorical categorical continuous class

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

One of the attributes is the class attribute
In this case: Cheat

Two class labels (or classes): Yes (1), No (0)

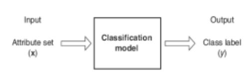

Figure 4.2. Classification as the task of mapping an input attribute set x into its class label y.

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What is classification (cont…)

- The target function *f* is known as a classification model
- Descriptive modeling: Explanatory tool to distinguish between objects of different classes (e.g., understand why people cheat on their taxes)
- Predictive modeling: Predict a class of a previously unseen record

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Examples of Classification Tasks

- Predicting tumor cells as benign or malignant
- Classifying credit card transactions as legitimate or fraudulent
- Categorizing news stories as finance, weather, entertainment, sports, etc
- Identifying spam email, spam web pages, adult content
- Understanding if a web query has commercial intent or not

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General approach to classification

- Training set consists of records with known class labels
- Training set is used to build a classification model
- A labeled test set of previously unseen data records is used to evaluate the quality of the model.
- The classification model is applied to new records with unknown class labels

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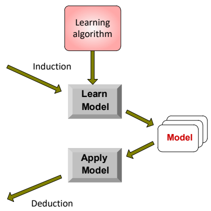
Illustrating Classification Task

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set



Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set

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Evaluation of classification models

- Counts of test records that are correctly (or incorrectly) predicted by the classification model
- **Confusion matrix**

		Predicted Class	
		Class = 1	Class = 0
Actual Class	Class = 1	f_{11}	f_{10}
	Class = 0	f_{01}	f_{00}

$$\text{Accuracy} = \frac{\#\,\text{correct predictions}}{\text{total}\,\#\,\text{of predictions}} = \frac{f_{11} + f_{00}}{f_{11} + f_{10} + f_{01} + f_{00}}$$

$$\text{Error rate} = \frac{\#\,\text{wrong predictions}}{\text{total}\,\#\,\text{of predictions}} = \frac{f_{10} + f_{01}}{f_{11} + f_{10} + f_{01} + f_{00}}$$

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Classification Techniques

- Decision Tree based Methods
- Rule-based Methods
- Memory based reasoning
- Neural Networks
- Naïve Bayes and Bayesian Belief Networks
- Support Vector Machines

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- Support Vector Machines

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Decision Trees

- Decision tree
  - A flow-chart-like tree structure
  - Internal node denotes a test on an attribute
  - Branch represents an outcome of the test
  - Leaf nodes represent class labels or class distribution

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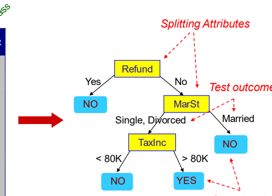
Example of a Decision Tree

categorical categorical continuous class

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Training Data



Model: Decision Tree

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Another Example of Decision Tree

categorical categorical continuous class

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
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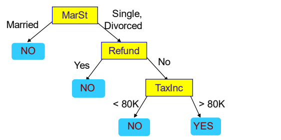
10	No	Single	90K	Yes



There could be more than one tree that fits the same data!

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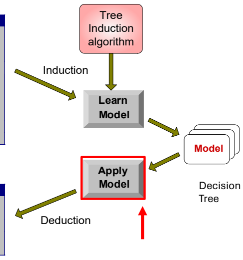
Decision Tree Classification Task

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



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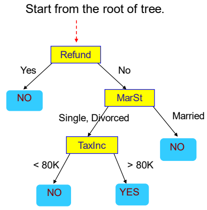
Apply Model to Test Data

Start from the root of tree.



Test Data

Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?

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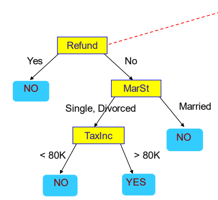
Apply Model to Test Data



Test Data

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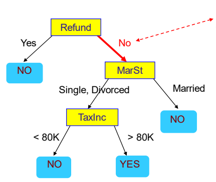
Apply Model to Test Data



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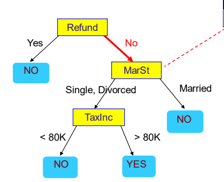

Test Data

Refund	Marital Status	Taxable Income	Cheat
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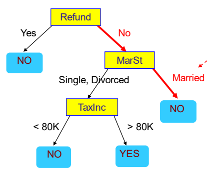
Apply Model to Test Data

Test Data

Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



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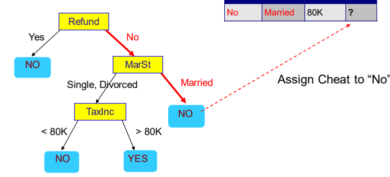
Apply Model to Test Data

Test Data

Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



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Decision Tree Classification Task

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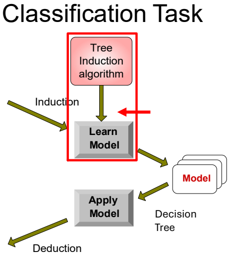
Training Set



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Test Set

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Tree Induction

- Finding the best decision tree is NP-hard
- Greedy strategy.
  - Split the records based on an attribute test that optimizes certain criterion.
- Many Algorithms:
  - Hunt's Algorithm (one of the earliest)
  - CART
  - ID3, C4.5
  - SLIQ, SPRINT

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General Structure of Hunt's Algorithm

- Let D_t be the set of training records that reach a node t
- General Procedure:
  - If D_t contains records that belong the same class y_t, then t is a leaf node labeled as y_t
  - If D_t contains records with the same attribute values, then t is a leaf node labeled with the majority class y_t
  - If D_t is an empty set, then t is a leaf node labeled by the default class, y_d
  - If D_t contains records that belong to more than one class, use an attribute test to split the data into smaller subsets.
    - Recursively apply the procedure to each subset.

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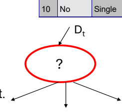

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Tree Induction

- Issues
  - How to **Classify** a leaf node
    - Assign the majority class
    - If leaf is empty, assign the default class – the class that has the highest popularity.
  - Determine how to split the records
    - How to specify the attribute test condition?
    - How to determine the best split?
  - Determine when to stop splitting

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How to Specify Test Condition?

- Depends on attribute types
  - Nominal
  - Ordinal
  - Continuous
- Depends on number of ways to split
  - 2-way split
  - Multi-way split

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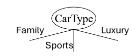
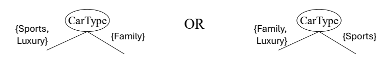
Splitting Based on Nominal Attributes

- Multi-way split: Use as many partitions as distinct values.



- Binary split: Divides values into two subsets. Need to find optimal partitioning.



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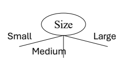
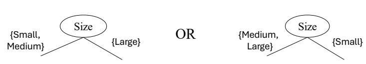
Splitting Based on Ordinal Attributes

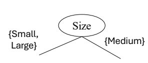
- Multi-way split: Use as many partitions as distinct values.



- Binary split: Divides values into two subsets – respects the order. Need to find optimal partitioning.



- What about this split?



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Splitting Based on Continuous Attributes

- Different ways of handling
  - Discretization to form an ordinal categorical attribute
    - Static – discretize once at the beginning
    - Dynamic – ranges can be found by equal interval bucketing, equal frequency bucketing (percentiles), or clustering.
  - Binary Decision: $(A < v)$ or $(A \geq v)$
    - consider all possible splits and finds the best cut
    - can be more compute intensive

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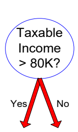
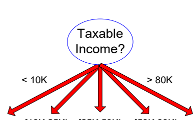
Splitting Based on Continuous Attributes



(i) Binary split



(ii) Multi-way split

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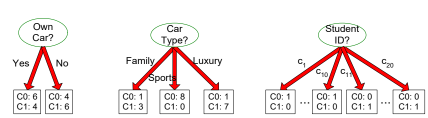
How to determine the Best Split

Before Splitting: 10 records of class 0,
10 records of class 1



Which test condition is the best?

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Which Attribute is the Best Classifier?

The central choice in the **ID3 algorithm** is selecting which attribute to test at each node in the tree

We would like to select the attribute which is most useful for classifying examples

For this we need a good quantitative measure

For this purpose a statistical property, called *information gain* is used

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Which Attribute is the Best Classifier?
Definition of **Entropy**

- In order to define information gain precisely, we begin by defining entropy
- Entropy is a measure commonly used in information theory.
- *Entropy characterizes the impurity of an arbitrary collection of examples*

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Entropy

- Entropy (D)
  - Entropy of data set D is denoted by H(D)
  - C_is are the possible classes
  - p_i = fraction of records from D that have class C

$$H(D) = -\sum_{c_i} p_i \log_2 p_i$$

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Entropy Examples

- Example:
  - 10 records have class A
  - 20 records have class B
  - 30 records have class C
  - 40 records have class D
  - Entropy = -[(.1 log .1) + (.2 log .2) + (.3 log .3) + (.4 log .4)]
  - Entropy = 1.846

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Splitting Criterion

- Example:
  - Two classes, +/-
  - 100 records overall (50 +s and 50 -s)
  - A and B are two binary attributes
    - Records with A=0: 48+, 2-
      Records with A=1: 2+, 48-
    - Records with B=0: 26+, 24-
      Records with B=1: 24+, 26-
  - Splitting on A is better than splitting on B
    - A does a good job of separating +s and -s
    - B does a poor job of separating +s and -s

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Which Attribute is the Best Classifier?
Information Gain

The expected information needed to classify a tuple in D is

$$Info(D) = -\sum_{i=1}^{m} p_i \log_2(p_i),$$ = Entropy

How much more information would we still need (after partitioning at attribute A) to arrive at an exact classification? This amount is measured by

$$Info_A(D) = \sum_{j=1}^{v} \frac{|D_j|}{|D|} \times Info(D_j).$$ = H(D, A)

Info Gain (D, A) = H(D) – H(D, A)

In general, we write Gain (D, A), where D is the collection of examples & A is an attribute

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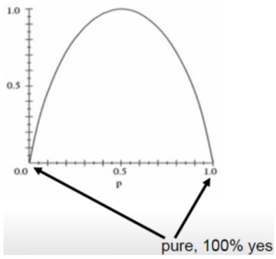
Entropy



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Information Gain

- ***Gain of an attribute split:*** compare the impurity of the parent node with the average impurity of the child nodes

$$InfoGain = H(parent) - \sum_{j=1}^{v} \frac{D_j}{D} H(D_j)$$

- Maximizing the gain ⇔ Minimizing the weighted average impurity measure of children nodes

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Examples Constructing Decision Tree

Class-Labeled Training Tuples from the *AllElectronics* Customer Database

RID	age	income	student	credit_rating	Class: buys_computer
1	youth	high	no	fair	no
2	youth	high	no	excellent	no
3	middle_aged	high	no	fair	yes
4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle_aged	low	yes	excellent	yes
8	youth	medium	no	fair	no
9	youth	low	yes	fair	yes
10	senior	medium	yes	fair	yes
11	youth	medium	yes	excellent	yes
12	middle_aged	medium	no	excellent	yes
13	middle_aged	high	yes	fair	yes
14	senior	medium	no	excellent	no

$Log_2(16) = log_{10}(16)/log_{10}(2) = 4$

$$\frac{log_c b}{log_c a}$$

$$Info(D) = -\frac{9}{14}\log_2\left(\frac{9}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right) = 0.940 \text{ bits.}$$

$$Info_{age}(D) = \frac{5}{14} \times \left(-\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5}\right) + \frac{4}{14} \times \left(\frac{4}{4}\log_2\frac{4}{4}\right) + \frac{5}{14} \times \left(-\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5}\right)$$
$$= 0.694 \text{ bits.}$$

$$Gain(age) = Info(D) - Info_{age}(D) = 0.940 - 0.694 = 0.246 \text{ bits.}$$

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Examples Constructing Decision Tree

$Gain(income) = 0.029$ bits
$Gain(student) = 0.151$ bits,
$Gain(credit\_rating) = 0.048$ bits

- So the attribute Age will be placed at root level.
- For placement at second level we find InfoGain for all the remaining attributes under every branch of the parent node.

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DECISION TREES

Which Attribute is the Best Classifier?: Information Gain

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

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DECISION TREES

Which Attribute is the Best Classifier?: Information Gain

The collection of examples has 9 positive values and 5 negative ones

$$Entropy(S) = 0.940$$

Eight (6 positive and 2 negative ones) of these examples have the attribute value *Wind = Weak*

Six (3 positive and 3 negative ones) of these examples have the attribute value *Wind = Strong*

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a) Entropy using the frequency table of one attribute:

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$E(S) = \sum_{i=1}^{c} - p_i \log_2 p_i$$

Play Golf	
Yes	No
9	5

Entropy(PlayGolf) = Entropy (5,9)
= Entropy (0.36, 0.64)
= - (0.36 $\log_2$ 0.36) - (0.64 $\log_2$ 0.64)
= 0.94

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b) Entropy using the frequency table of two attributes: (Outlook)

$$E(T,X) = \sum_{c \in X} P(c)E(c)$$

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

		Play Golf		
		Yes	No	
Outlook	Sunny	3	2	5
	Overcast	4	0	4
	Rainy	2	3	5
				14

E(PlayGolf, Outlook) = P(Sunny)*E(3,2) + P(Overcast)*E(4,0) + P(Rainy)*E(2,3)
= (5/14)*0.971 + (4/14)*0.0 + (5/14)*0.971
= 0.693

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Information Gain

$$Gain(T,X) = Entropy(T) - Entropy(T,X)$$

G(PlayGolf, Outlook) = E(PlayGolf) – E(PlayGolf, Outlook)
= 0.940 – 0.693 = 0.247

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		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3
Gain = 0.247			

		Play Golf	
		Yes	No
Temp.	Hot	2	2
	Mild	4	2
	Cool	3	1
Gain = 0.029			

		Play Golf	
		Yes	No
Humidity	High	3	4
	Normal	6	1
Gain = 0.152			

		Play Golf	
		Yes	No
Windy	False	6	2
	True	3	3
Gain = 0.048			

Step 3: Choose attribute with the largest information gain as the decision node, divide the dataset by its branches and repeat the same process on every branch.

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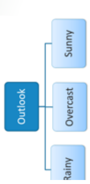
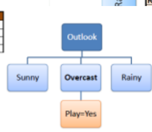
★		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3
Gain = 0.247			



Outlook	Temp	Humidity	Windy	Play Golf
Sunny	Mild	High	FALSE	Yes
Sunny	Cool	Normal	FALSE	Yes
Sunny	Cool	Normal	TRUE	No
Sunny	Mild	Normal	FALSE	Yes
Sunny	Mild	High	TRUE	No
Overcast	Hot	High	FALSE	Yes
Overcast	Cool	Normal	TRUE	Yes
Overcast	Mild	High	TRUE	Yes
Overcast	Hot	Normal	FALSE	Yes
Rainy	Hot	High	FALSE	No
Rainy	Hot	High	TRUE	No
Rainy	Mild	High	FALSE	No
ainy	Cool	Normal	FALSE	Yes
ainy	Mild	Normal	TRUE	Yes

Temp	Humidity	Windy	Play Golf
Hot	High	FALSE	Yes
Cool	Normal	TRUE	Yes
Mild	High	TRUE	Yes
Hot	Normal	FALSE	Yes



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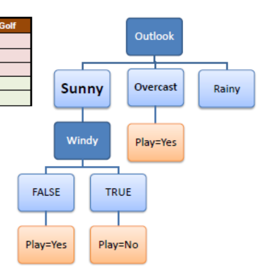
A branch with entropy more than 0 needs further splitting.

Temp	Humidity	Windy	Play Golf
Mild	High	FALSE	Yes
Cool	Normal	FALSE	Yes
Mild	Normal	FALSE	Yes
Cool	Normal	TRUE	No
Mild	High	TRUE	No



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Further development of Decision Tree

- Gain[$T_{(Sunny)}$, Temp]= E(T_{Sunny}) – E(T_{Sunny}, Temp)
- Gain[T_{Sunny}, Humidity] = E(T_{Sunny})- E(T_{Sunny}, Humidity)
- Gain[T_{Sunny}, Wind] = E(T_{Sunny})- E(T_{Sunny}, Wind)
- Choose the attribute with maximum gain for Sunny Table, after calculating, Humidity would have maximum gain.
- Similarly;
- Gain[T_{Rainy}, Temp]= E(T_{Rainy})- E(T_{Rainy}, Temp)
- Gain[T_{Rainy}, Wind]= E(T_{Rainy})- E(T_{Rainy}, Wind),
- No need to calculate the Gain[T_{Rain}, Humidity] because Humidity has already used in the decision tree by Sunny table.

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Decision Tree to Decision Rules

A decision tree can easily be transformed to a set of rules by mapping from the root node to the leaf nodes one by one.

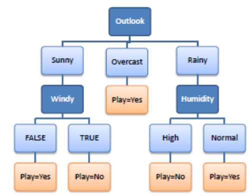
R_1: IF (Outlook=Sunny) AND (Windy=FALSE) THEN Play=Yes

R_2: IF (Outlook=Sunny) AND (Windy=TRUE) THEN Play=No

R_3: IF (Outlook=Overcast) THEN Play=Yes

R_4: IF (Outlook=Rainy) AND (Humidity=High) THEN Play=No

R_5: IF (Outlook=Rain) AND (Humidity=Normal) THEN Play=Yes



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DECISION TREES

Which Attribute is the Best Classifier?: Information Gain

The information gain obtained by separating the examples according to the attribute *Wind* is calculated as:

$$\begin{aligned}
Gain(S, Wind) &= Entropy(S) - \sum_{v \in \{Weak, Strong\}} \frac{|S_v|}{|S|} Entropy(S_v) \\
&= Entropy(S) - (8/14)Entropy(S_{Weak}) \\
&\quad - (6/14)Entropy(S_{Strong}) \\
&= 0.940 - (8/14)0.811 - (6/14)1.00 \\
&= 0.048
\end{aligned}$$

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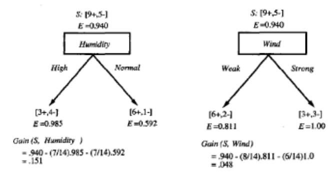
DECISION TREES

Which Attribute is the Best Classifier?: Information Gain

We calculate the Info Gain for each attribute and select the attribute having the highest Info Gain



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DECISION TREES

Example

Which attribute should be selected as the first test?

$$\begin{aligned}
Gain(S, Outlook) &= 0.246 \\
Gain(S, Humidity) &= 0.151 \\
Gain(S, Wind) &= 0.048 \\
Gain(S, Temperature) &= 0.029
\end{aligned}$$

"Outlook" provides the most information

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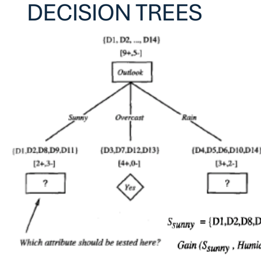
DECISION TREES



S_{sunny} = {D1,D2,D8,D9,D11}

$Gain(S_{sunny}, Humidity)$ = .970 - (3/5) 0.0 - (2/5) 0.0 = .970

$Gain(S_{sunny}, Temperature)$ = .970 - (2/5) 0.0 - (2/5) 1.0 - (1/5) 0.0 = .570

$Gain(S_{sunny}, Wind)$ = .970 - (2/5) 1.0 - (3/5) .918 = .019

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DECISION TREES

Example

The process of selecting a new attribute is now repeated for each (non-terminal) descendant node, this time using only training examples associated with that node

Attributes that have been incorporated higher in the tree are excluded, so that any given attribute can appear at most once along any path through the tree

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DECISION TREES

Example

This process continues for each new leaf node until either:

1. Every attribute has already been included along this path through the tree
2. The training examples associated with a leaf node have zero entropy

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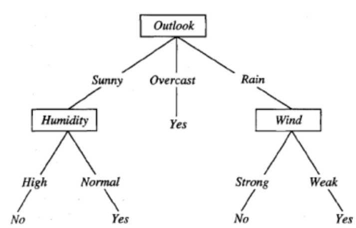
DECISION TREES

Example



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DECISION TREES

From Decision Trees to Rules

Next Step: Make rules from the decision tree

After making the identification tree, we trace each path from the root node to leaf node, recording the test outcomes as antecedents and the leaf node classification as the consequent

Simple way: one rule for each leaf

For our example we have:

If the Outlook is Sunny and the Humidity is High then No
If the Outlook is Sunny and the Humidity is Normal then Yes
...

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